SAMPLE

The Fine Structure Constant

The Mysterious Number 1/137

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Introduction

Every branch of physics keeps a small set of numbers that measure the world. The speed of light sets the conversion between space and time. The masses of the electron and the proton set the scale of matter. Newton's gravitational constant sets the strength of the weakest force in nature. But among all these numbers, one stands apart, because it is not a speed, a mass, or a strength at all. It has no units. It is a pure ratio, the same for every observer in every laboratory in the universe, and it can be written on a single line: the fine structure constant, approximately one divided by one hundred thirty-seven.

The constant was born in 1916, when the German physicist Arnold Sommerfeld extended Niels Bohr's early model of the atom to include the corrections demanded by Einstein's relativity. Bohr's model had explained the gross structure of the hydrogen spectrum, the broad separation of its colored lines. Sommerfeld's extension explained something finer: the observation, made in the closing years of the nineteenth century, that many of those lines were not single lines at all but closely spaced pairs and triplets, a pattern that physicists had named the fine structure of the spectrum. The number that set the spacing of these delicate sub-lines was a ratio of the electron's charge, the speed of light, and the new constant of action that Max Planck had introduced. Sommerfeld named it the fine structure constant and gave it the symbol of the first letter of the Greek alphabet.

That symbol turned out to be fitting in ways Sommerfeld could not have foreseen. In the century that followed, the constant outgrew the spectral lines that gave it its name. It became the coupling strength of the electromagnetic force itself, the number that tells how strongly electric charge interacts with light. It sets the size of atoms, the stiffness of chemical bonds, the melting point of iron, and the transparency of glass. It is the only fundamental constant of physics that can be arrived at without any knowledge of human units, and for that reason, it is the number that any hypothetical civilization beyond Earth would recognize as we do.

The constant is also, famously, a mystery. No theory in existence predicts its value. It cannot be derived from deeper principles; it must be measured, measured again, and then measured once more, to ever greater precision, in the hope that its value will one day reveal the reason for its size. Richard Feynman, one of the architects of the theory that explains electromagnetism better than any theory in science explains anything, called it one of the greatest damn mysteries of physics, a magic number delivered to humanity without an accompanying explanation.

This book is an account of that number: where it came from, what it measures, how it has been hunted to extraordinary precision, and why its value may hold a key to questions that extend beyond physics into chemistry, biology, and the conditions required for life itself. The chapters move from the spectroscopes of the nineteenth century through the quantum revolution, into the theory of light and matter that made the constant a pillar of modern science, and finally to the experiments that ask whether the number has remained the same across the history of the universe, and what would have happened if it had not.

Chapter 1: The Discovery of The Fine Structure Constant

The story begins with a puzzle hidden inside light itself. In 1887, the American physicist Albert Michelson, the same experimental genius who would later help prove the constancy of the speed of light, turned the resolving power of his precision instruments on the spectrum of hydrogen. Hydrogen was the favorite subject of spectroscopists because it is the simplest atom, and its spectrum, when dispersed through a prism or grating, resolves into a ladder of sharply defined lines whose positions follow such tidy mathematical patterns that physicists were convinced the lines carried a message about the atom's inner architecture. Michelson found that the red line of the hydrogen spectrum, one of the most carefully studied lines in all of physics, was not a single line at all. Under sufficient magnification it resolved into two lines, pressed so close together that no instrument of an earlier generation had ever separated them.

The puzzle mattered more than its discoverers could have guessed. Spectral lines are the atom's signature, and their fine detail promised to be the atom's testimony about its inner architecture. In the final decade of the nineteenth century, physicists were still debating whether atoms were real at all; many eminent scientists regarded them as useful fictions, bookkeeping devices for chemistry. Yet here were the fictions, if such they were, printing delicate doublets into the light of glowing hydrogen gas with the precision of an engraver. Whatever produced that pattern, it was small, it was lawful, and it was beyond every mechanism classical physics could offer. The fine structure joined a short list of stubborn facts, alongside the spectrum of thermal radiation and the photoelectric effect, that would not fit the classical picture and waited for the new physics that arrived in the first decades of the

twentieth century to be understood.

This splitting was not confined to hydrogen. As instruments improved through the 1890s and the first years of the twentieth century, spectroscopists found that many spectral lines, in many elements, were really clusters of lines. The pattern of these clusters came to be called the fine structure of the spectrum. The name was apt: the splitting was fine, a matter of small wavelength differences, and it was structural, suggesting that something inside the atom had a hidden architecture of its own. Classical physics could not explain it. A charged particle moving in a circle radiates light, and its orbit should decay; a stable atom of the classical kind could not exist in the first place, let alone one that produced delicate pairs of spectral lines.

The first breakthrough came in 1913, when Niels Bohr proposed his quantum model of the atom. Bohr postulated that the electron in hydrogen could occupy only certain allowed orbits, each with a definite energy, and that spectral lines were emitted when the electron dropped from one allowed orbit to another. The model was astonishingly successful with the coarse spectrum: it reproduced the observed wavelengths of the visible hydrogen lines to within the accuracy of measurement. But Bohr's model, in its original form, had every orbit of a given energy on exactly one shelf. It predicted single lines where Michelson had found doubles. The fine structure was beyond its reach.

The missing ingredient came from Einstein's special theory of relativity. Bohr's electron moved around the nucleus at speeds that, though modest compared with light, were fast enough that the electron's mass shifted measurably from orbit to orbit. In 1916, Arnold Sommerfeld, a professor in Munich who ran one of the great schools of theoretical physics in the world, set out to repair the omission. Sommerfeld allowed the electron to travel not in circles but in ellipses, and he applied the relativistic correction to the electron's motion. When he did so, the single shelves of Bohr's model split. Each old energy level separated into a cluster of closely spaced levels, and each old spectral line became a family of lines pressed close together. The pattern was exactly the fine structure the spectroscopists had been mapping for thirty years.

Sommerfeld's analysis produced a ratio of three quantities already known to physics: the charge of the electron, the speed of light, and Planck's constant of action, together with the electric constant of the vacuum. The combination came out as a pure number, very nearly one over one hundred thirty-seven. It measured, in his formula, the relative size of the relativistic correction, and therefore the spacing of the fine structure lines. Sommerfeld gave the number both its name and its symbol, the first letter of the Greek alphabet. In the atomic world it played the role of a small parameter: the smaller the splitting, the more precise the instrument needed to see it, and the more beautiful the atomic spectrum revealed itself to be.

The discovery made Sommerfeld's name inseparable from the constant, though his relativity-based derivation was not the final word. Sommerfeld's elliptical orbits, beautiful as the mathematics was, would eventually be replaced by the quantum mechanics of the mid-1920s, and the quantum mechanics in turn would be refined by the quantum theory of fields. But through every reconstruction, the number itself survived unchanged. Later theory would show that alpha is not primarily a fact about hydrogen at all. It is the coupling constant of the electromagnetic interaction, the single number that determines how strongly any charged particle anywhere in the universe talks to light. Sommerfeld had pulled a thread that would unravel into a century of physics.

The historical context deserves a final note. Sommerfeld never won a Nobel Prize, though he was nominated more times than perhaps any physicist of his stature, and though his students included Werner Heisenberg and Wolfgang Pauli, two of the founders of the quantum theory that built on his work. His textbook on atomic structure and spectral lines trained a generation of physicists in the years when the quantum was new. And his constant outlived the model that produced it. When physics moved on, alpha remained, one of the few numbers of the old quantum theory to pass intact into the new one, and the only one to carry with it a mystery that is still open today.